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Optical coherence tomography imaging in non-destructive testing applications

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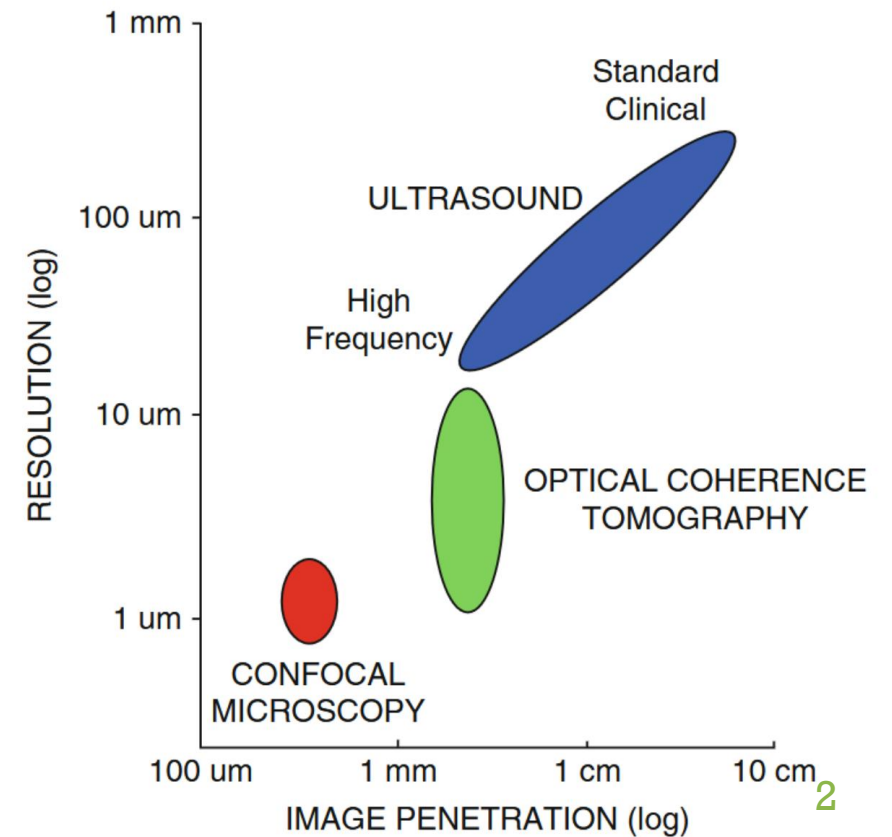
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INTRODUCTION: OCT

- OCT is an optical non-destructive and non-invasive imaging method, which allows high resolution acquisition of one-, two- or three-dimensional depth-resolved data;
- Fills a niche between ultrasound and confocal microscopy when it comes to resolution and image depth needs;

Ultrasound and OCT can be considered analogous:

- A beam of sound/light is incident into the sample
- Reflection/scattering happens due to structures with varying acoustic/optical properties
- These internal structures can be determined by measuring the “echo” time it takes for sound/light to travel different axial distances.



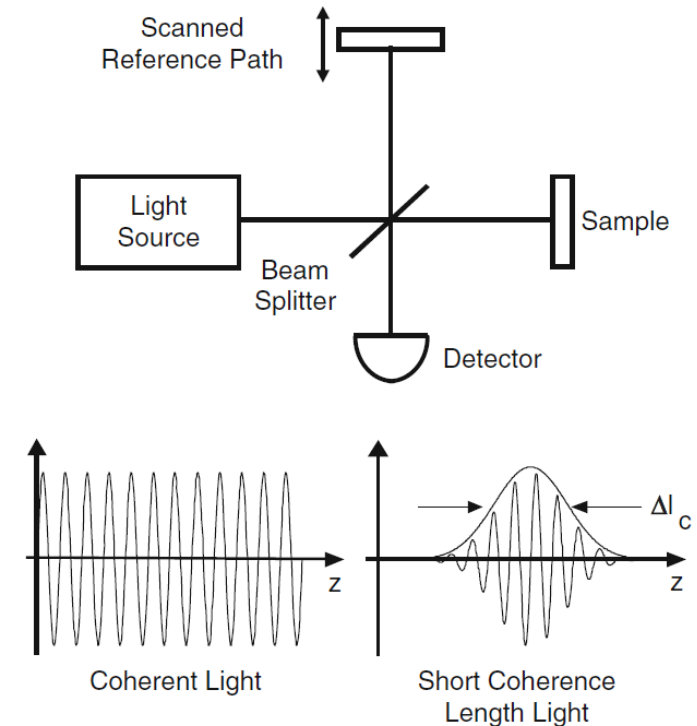
OCT VS ULTRASOUND

However:

- The speed of sound is 1,500 m/s, while the speed of light is approximately 3×10^8 m/s.
- In order to measure distances with a $100 \mu\text{m}$ resolution, a typical resolution for ultrasound imaging, a time resolution of 100 ns is required. This resolution is well within the limits of electronic detection
- The measurement of distances with a $10 \mu\text{m}$ resolution, a typical resolution for OCT imaging, requires a time resolution of 30 fs (30×10^{-15} s).
→ direct measurement is impossible.

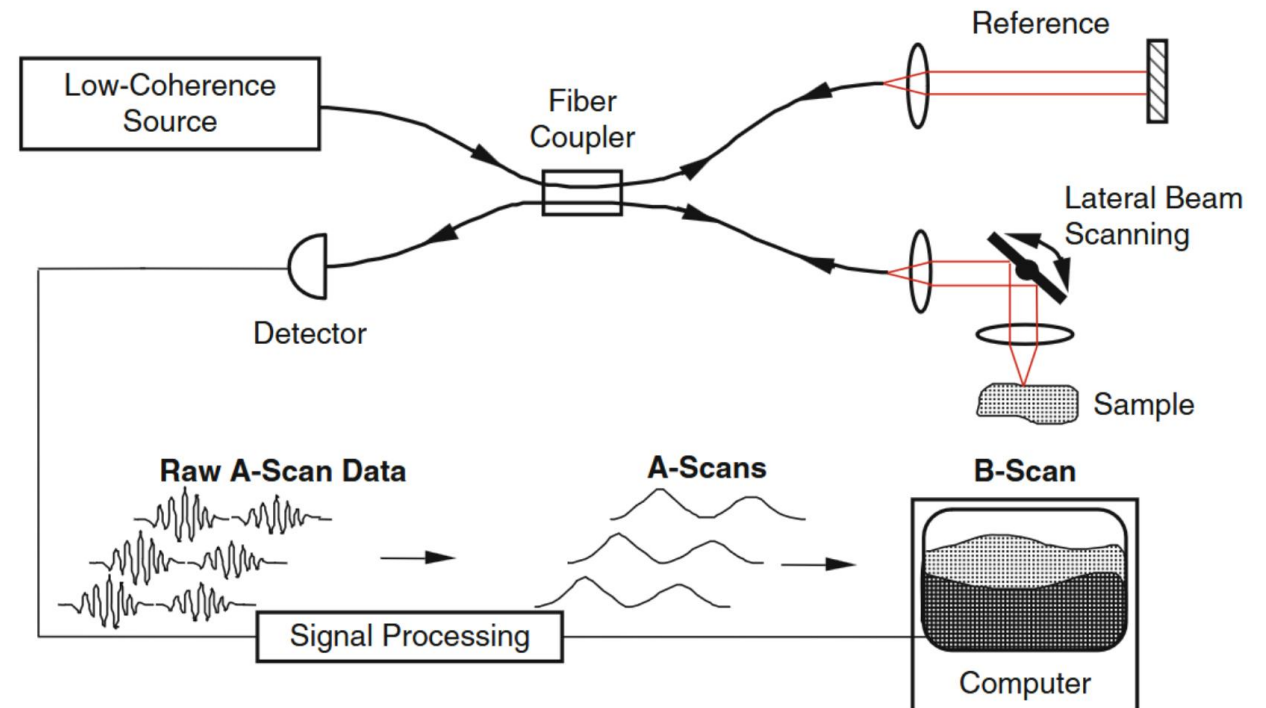
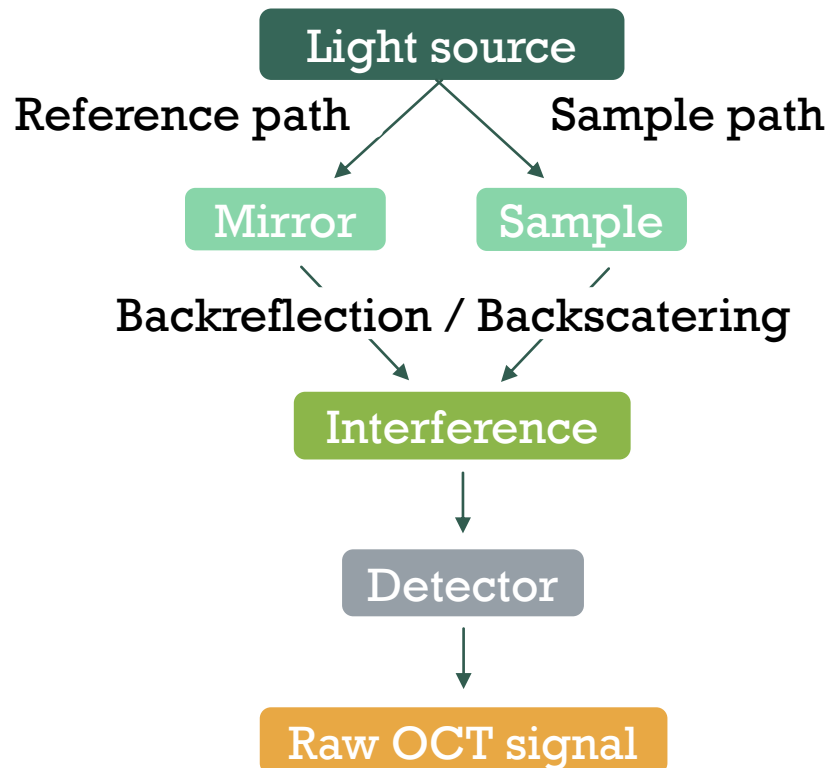
Solution: interferometry

Classic Michelson interferometer

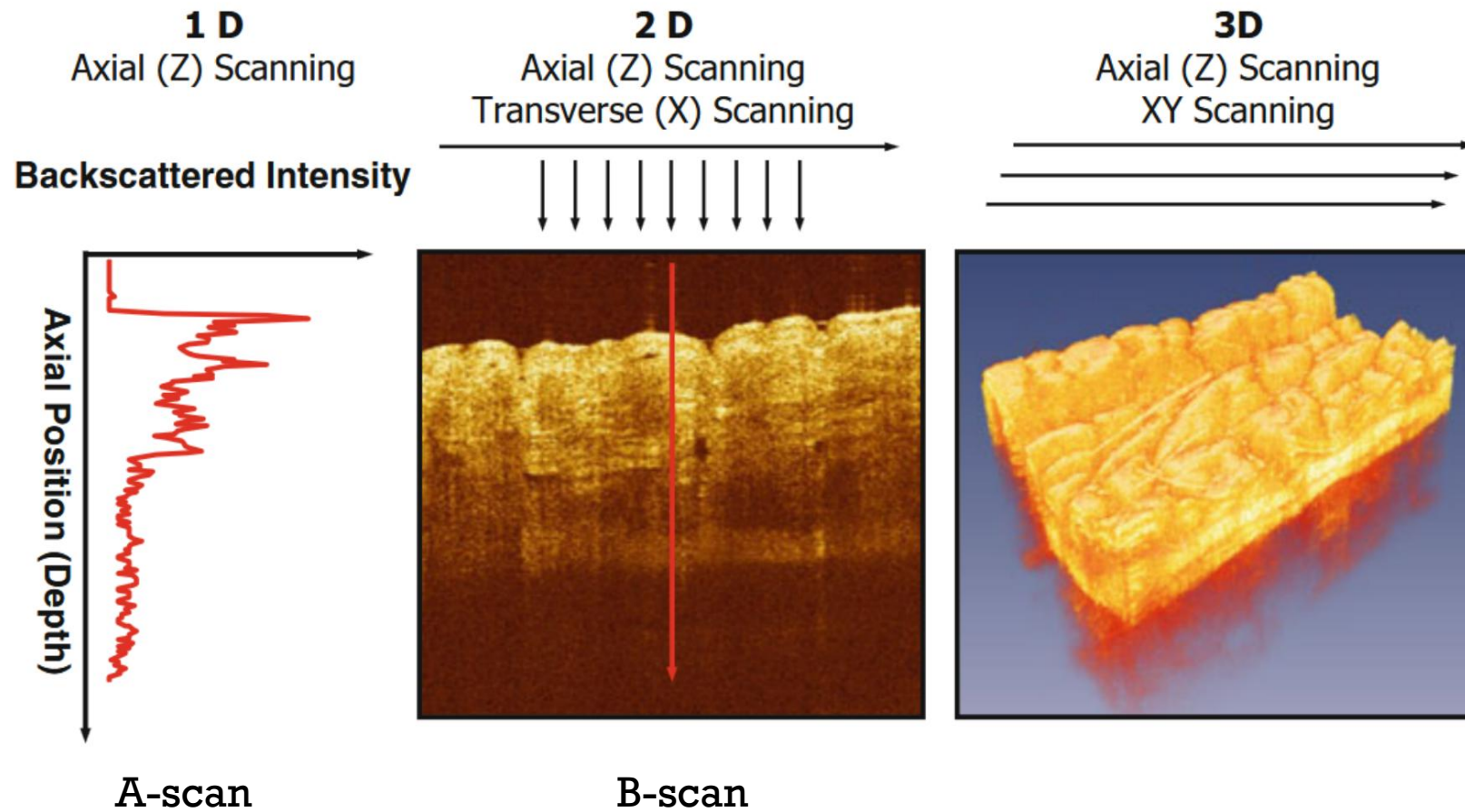


PRINCIPLE OF OCT

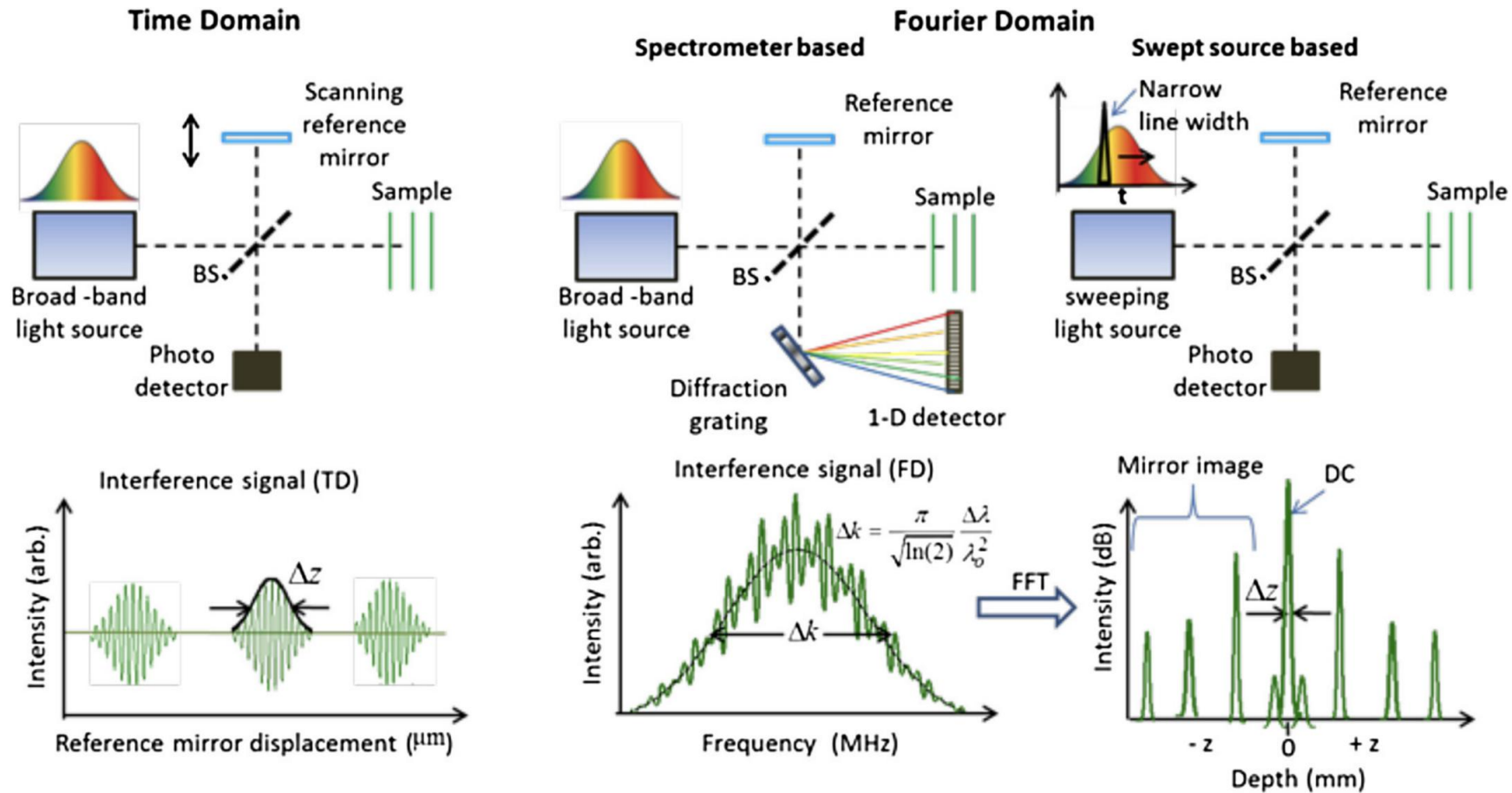
- OCT is a non-contact sensing tool that operates based on the principle of low coherence interferometry and relies on time delays and magnitude of light echoes to generate images with microscopic resolution.



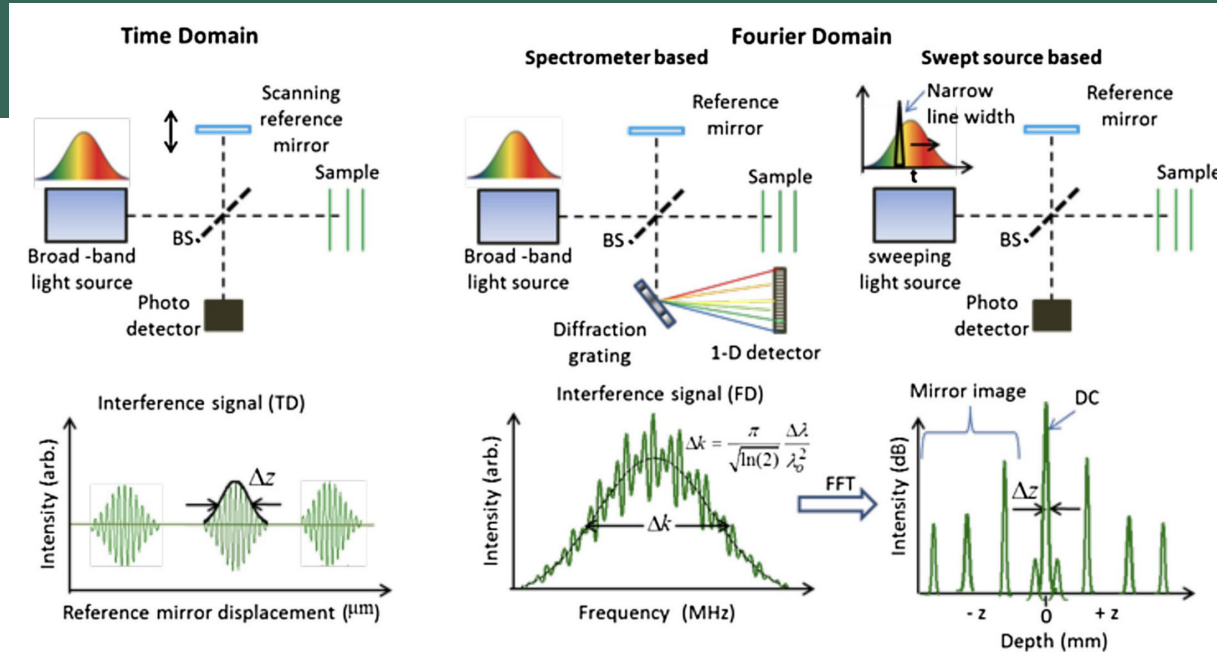
PRINCIPLE OF OCT



PRINCIPLE OF OCT: TIME VS FOURIER DOMAIN



PRINCIPLE OF OCT: TIME VS FOURIER DOMAIN



OCT Type	Method of Depth Encoding	Advantages	Limitations
TD-OCT	Moving reference mirror	Simpler design	Slow, lower sensitivity
SD-OCT	Spectrometer + detector array	Fast, high sensitivity	Reduced penetration in scattering tissues
SS-OCT	Swept laser + single detector	Deeper penetration, less roll-off	More complex and expensive

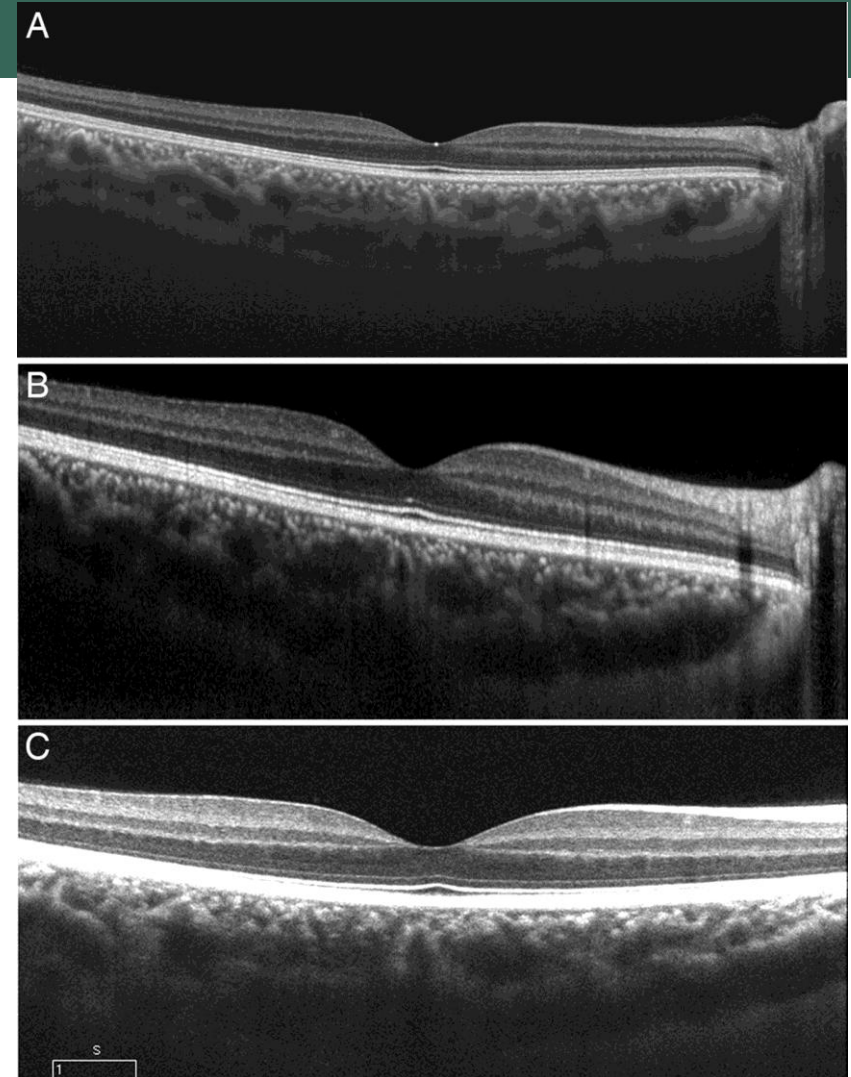
PERFORMANCE METRICS

Metric	Definition	Main Influencers
Axial Resolution	Ability to separate layers in depth	• Light source spectral bandwidth (wider is better) • Center wavelength (shorter is better) • Dispersion in optics or tissue
Lateral Resolution	Clarity of detail across (sideways) the beam	• Numerical aperture (NA) of focusing optics • Beam diameter / focusing spot size • Wavelength (indirectly - shorter usually gives finer lateral detail)
Penetration Depth	How far light can image inside tissue	• Absorption + scattering of the tissue • Wavelength choice (near-IR penetrates more) • Detector sensitivity & illumination power
Depth of Focus (DOF)	Range over which the lateral resolution holds	• NA of optics (high NA improves lateral but narrows DOF) • Wavelength & beam diameter entering optics

- In OCT, the axial and lateral properties are decoupled from each other.

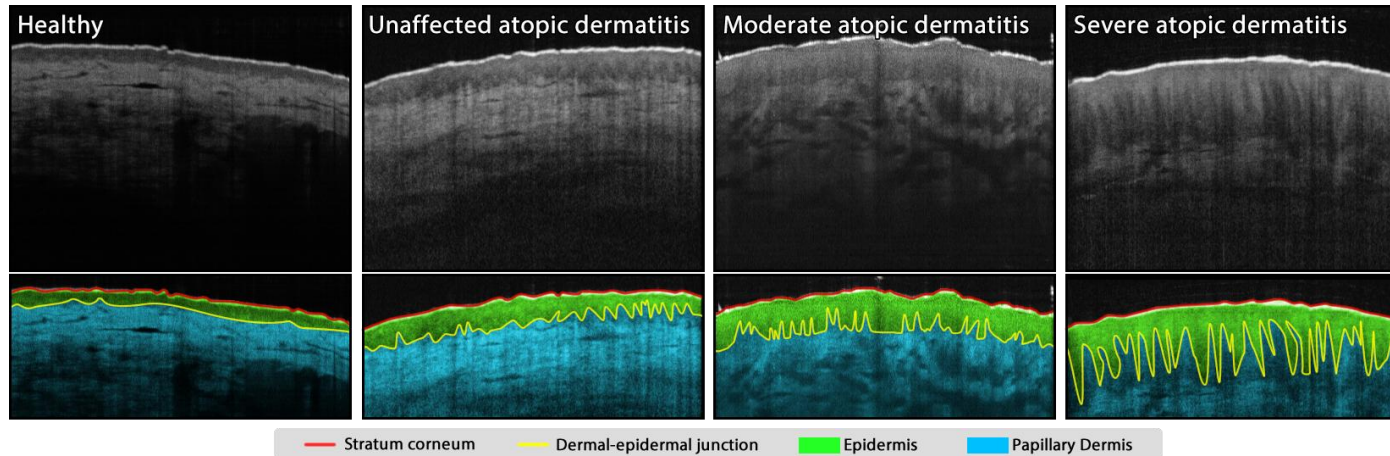
APPLICATIONS OF OCT

- Particularly prominent in the field of ophthalmology → fundamental diagnostic tool for detecting and monitoring eye diseases (such as glaucoma, diabetic retinopathy, and macular degeneration) by providing detailed measurements of retinal thickness and nerve fiber layer thickness.

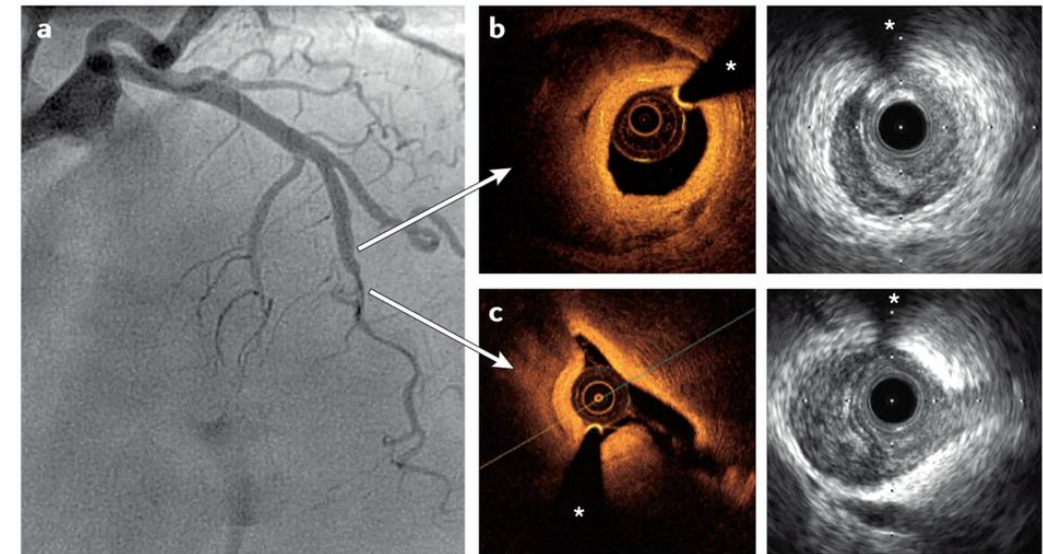


APPLICATIONS OF OCT

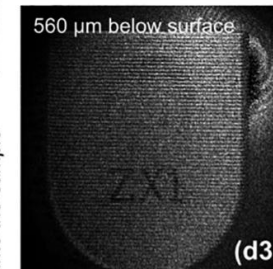
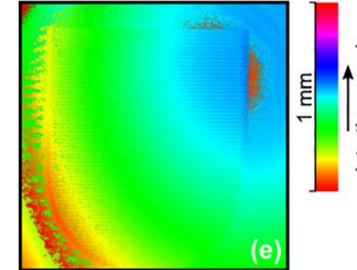
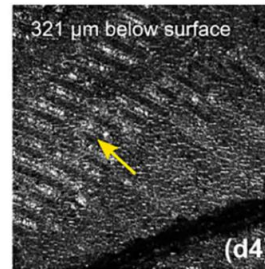
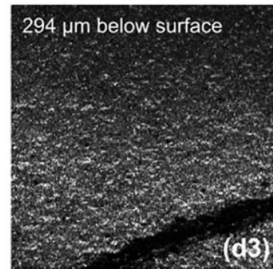
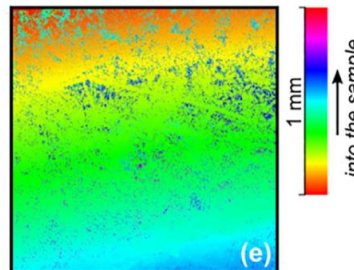
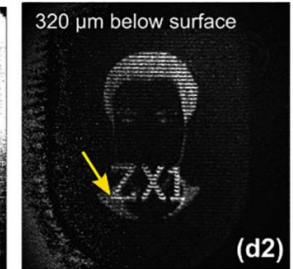
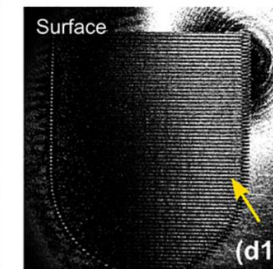
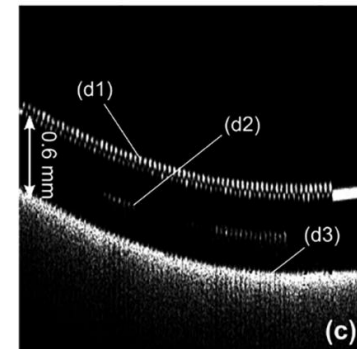
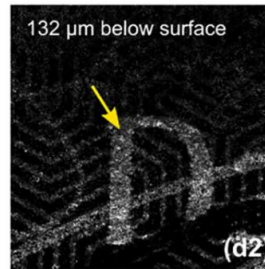
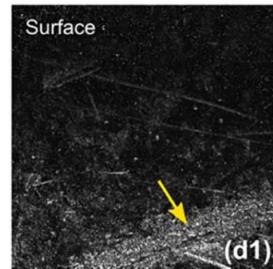
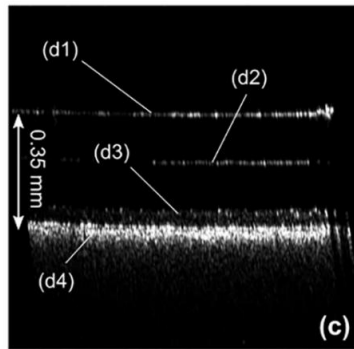
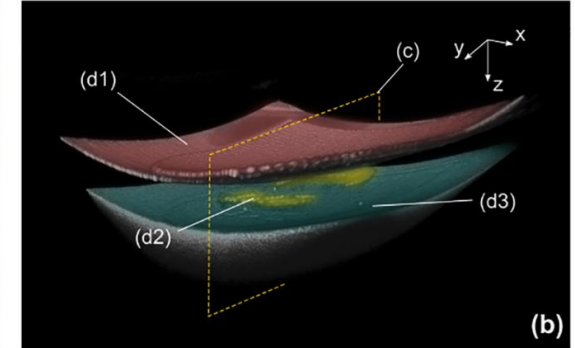
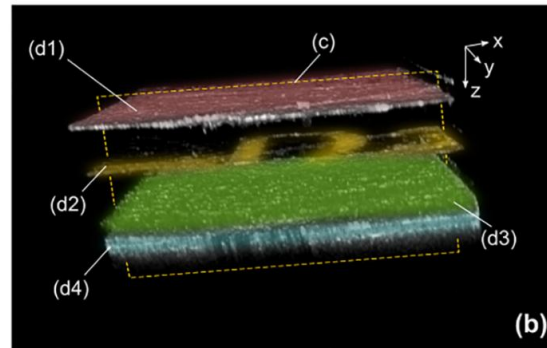
- More recently, OCT has been applied to many other medical specialties such as dermatology, cardiology, etc.



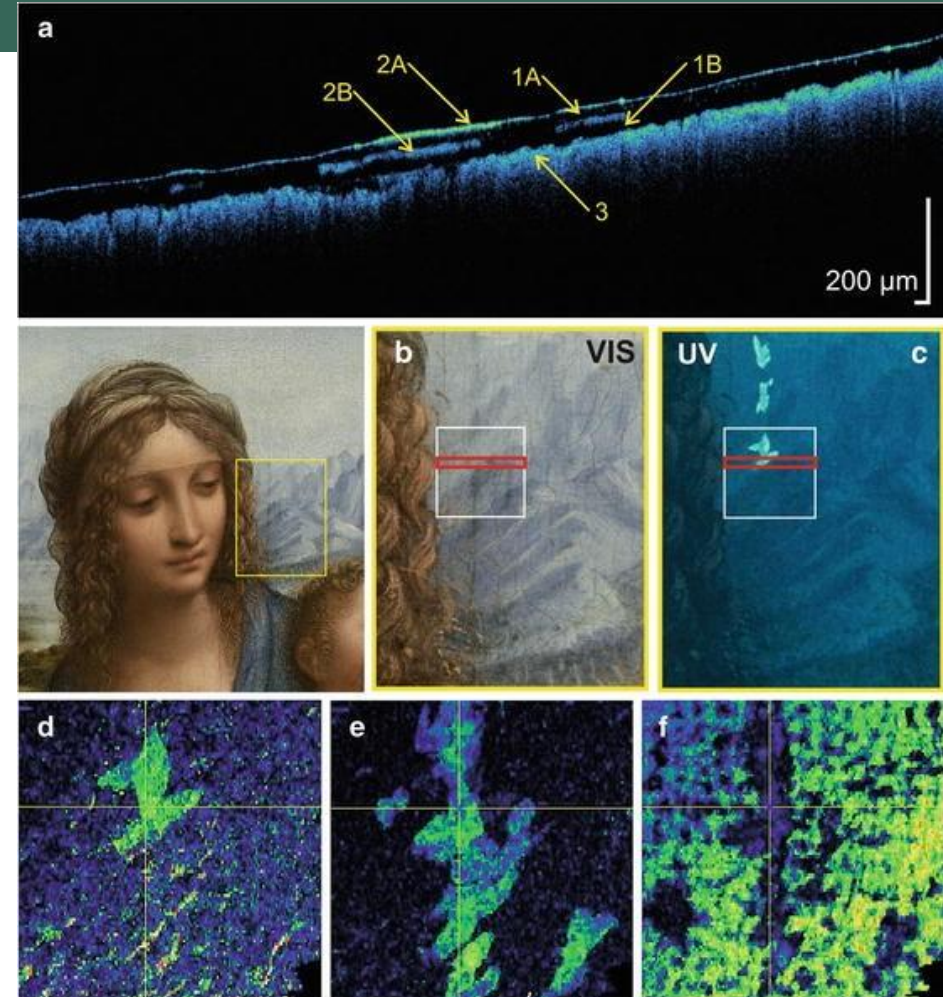
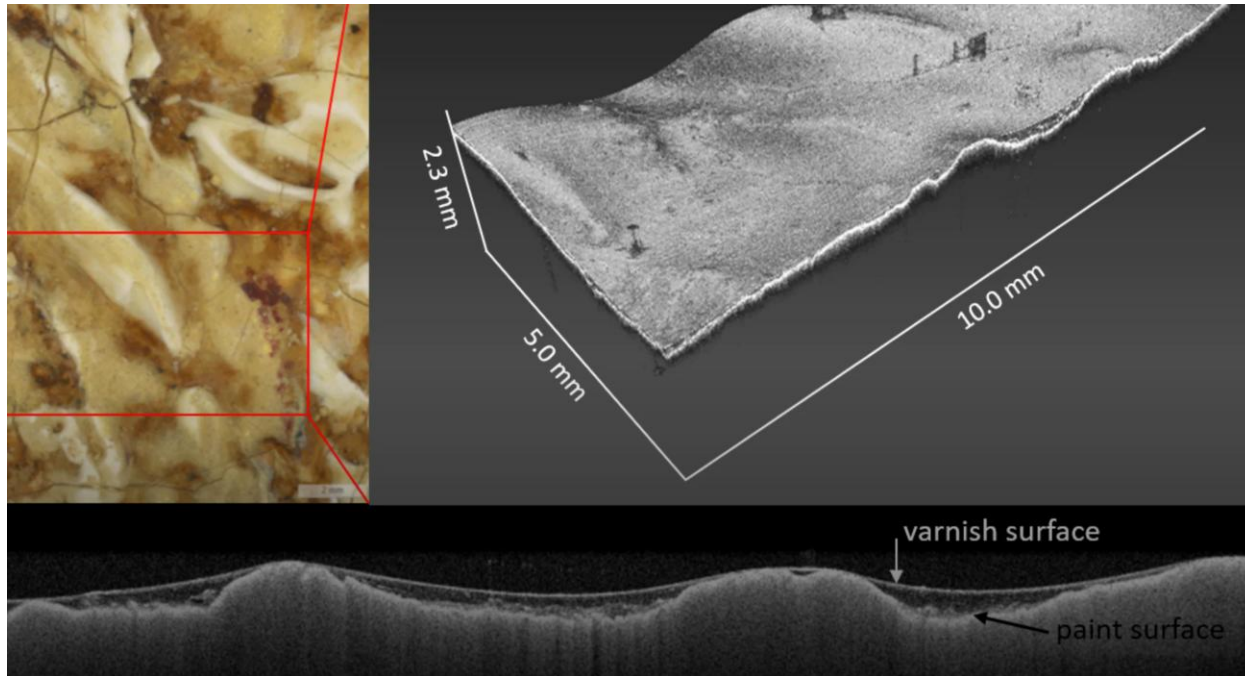
- However, non-medical applications are relatively underdeveloped due to high costs of equipment and maintenance, specific material optical properties and market interest. Nonetheless, its potencial has been shown in many diferente areas...



OCT IN FORENSIC SCIENCES: EXAMPLES

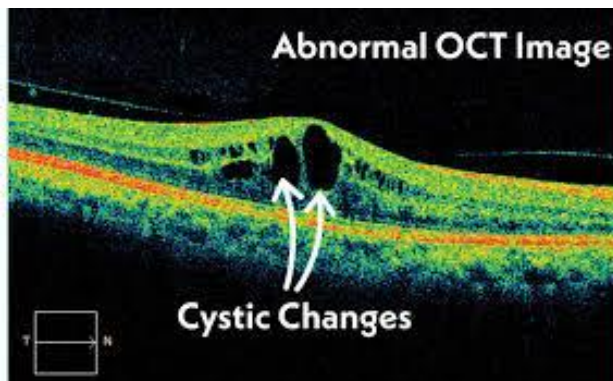
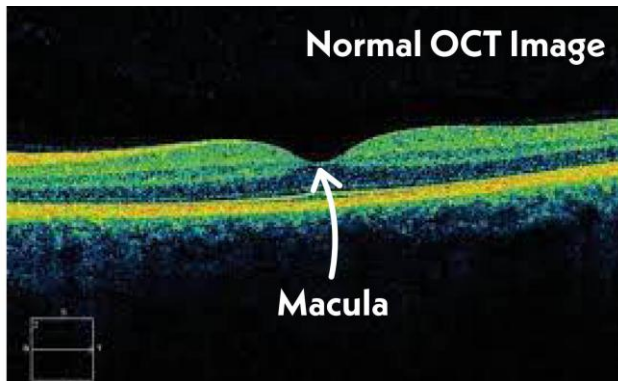


OCT IN ARTWORK INSPECTION: EXAMPLES



OCT IN SPACE

- Extensively used in ophthalmology to image the retina

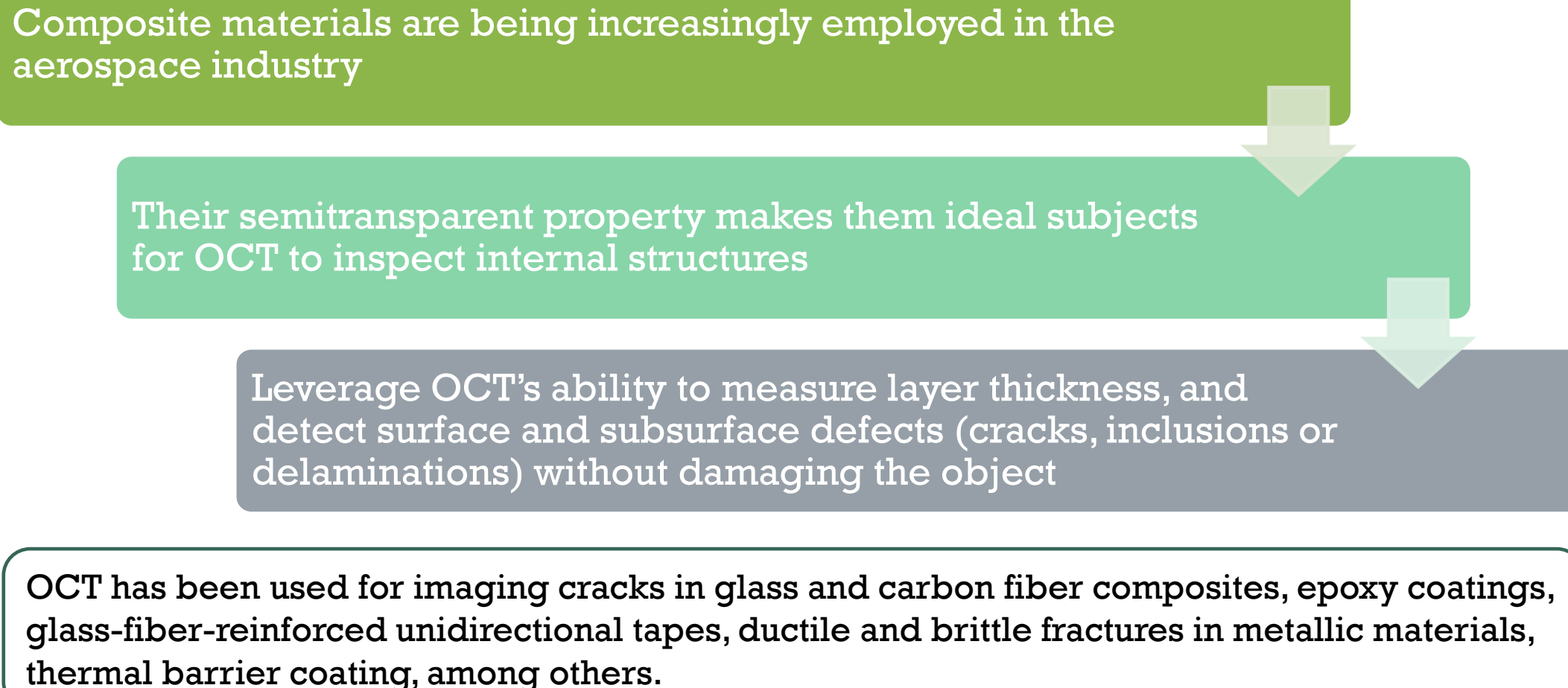


- Including astronauts in the **International Space Station**
 - Used to investigate the effects of a microgravity environment on vision - space flight associated neuro-ocular syndrome (SANS)



APPLICATION OF OCT IN AEROSPACE ENGINEERING

Composite materials are being increasingly employed in the aerospace industry



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graph TD; A[Composite materials are being increasingly employed in the aerospace industry] --> B[Their semitransparent property makes them ideal subjects for OCT to inspect internal structures]; B --> C[Leverage OCT's ability to measure layer thickness, and detect surface and subsurface defects (cracks, inclusions or delaminations) without damaging the object]; C --> D[OCT has been used for imaging cracks in glass and carbon fiber composites, epoxy coatings, glass-fiber-reinforced unidirectional tapes, ductile and brittle fractures in metallic materials, thermal barrier coating, among others.];
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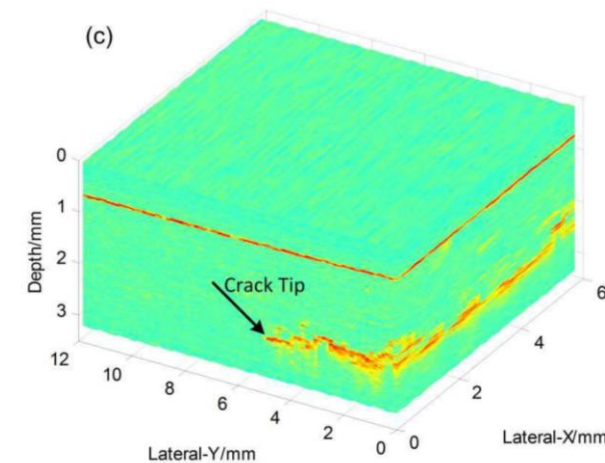
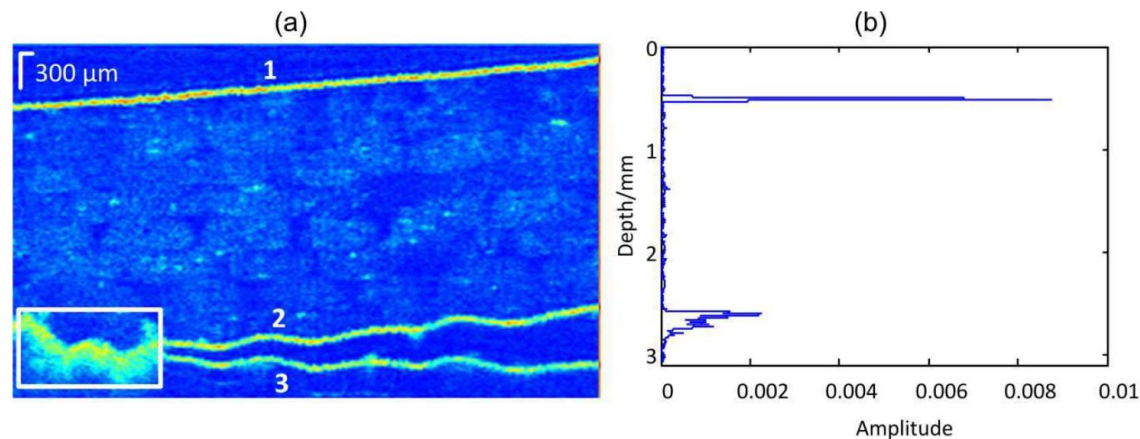
Their semitransparent property makes them ideal subjects for OCT to inspect internal structures

Leverage OCT's ability to measure layer thickness, and detect surface and subsurface defects (cracks, inclusions or delaminations) without damaging the object

OCT has been used for imaging cracks in glass and carbon fiber composites, epoxy coatings, glass-fiber-reinforced unidirectional tapes, ductile and brittle fractures in metallic materials, thermal barrier coating, among others.

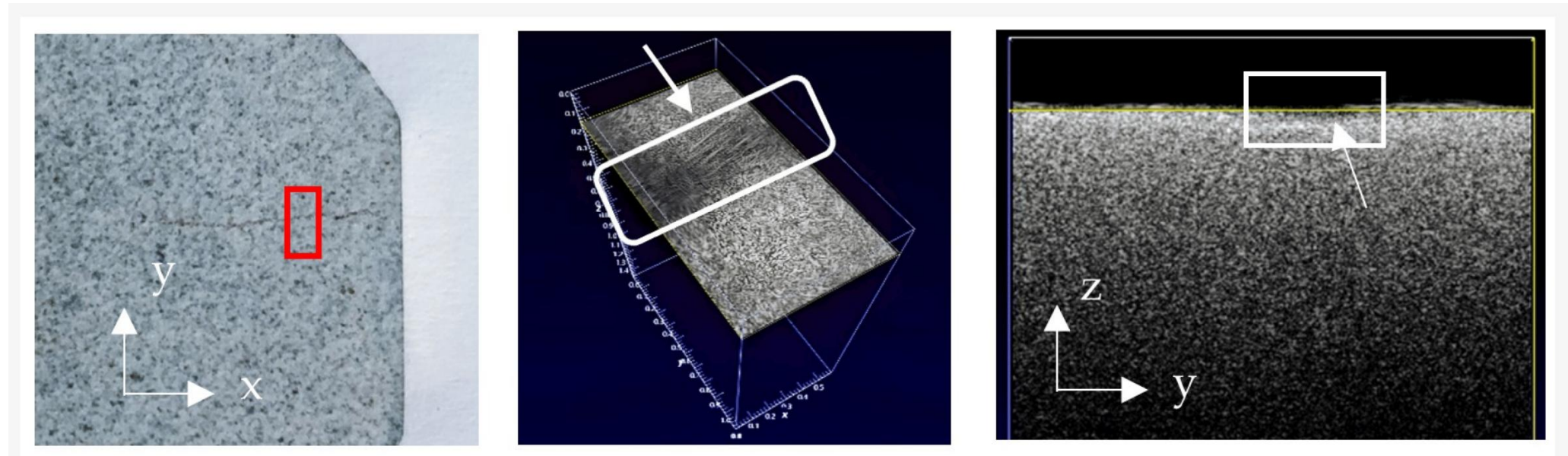
OCT IN AEROSPACE ENGINEERING: EXAMPLES

- The ability to measure layer thickness, and detect surface and subsurface defects without damaging the object is desired → materials may be reused and retested
- OCT can analyze **composites, which are gaining popularity** aerospace components → great physical properties, but challenging to evaluate structural integrity (fatigue cracks, impact damages, debonding and **delamination**)



OCT IN AEROSPACE ENGINEERING: EXAMPLES

- Also applied to study defects in thermal barrier coatings, often used in the engines of aircrafts



OCT IN AEROSPACE ENGINEERING: CHALLENGES

A potential limitation of OCT is its imaging depth, which is heavily constrained by the scattering of light through inhomogeneous samples.

Longer wavelengths could enhance it → Light sources and detectors become less efficient and there is a significant increase in absorption for many materials

A trade-off between imaging depth and reliability and ease of use needs to be achieved for each specific case.

OCT wavelength	Penetration Depth	Defect depth resolution
1.3 μm	10-100 μm	1-15 μm
4 μm	400 μm	7-30 μm

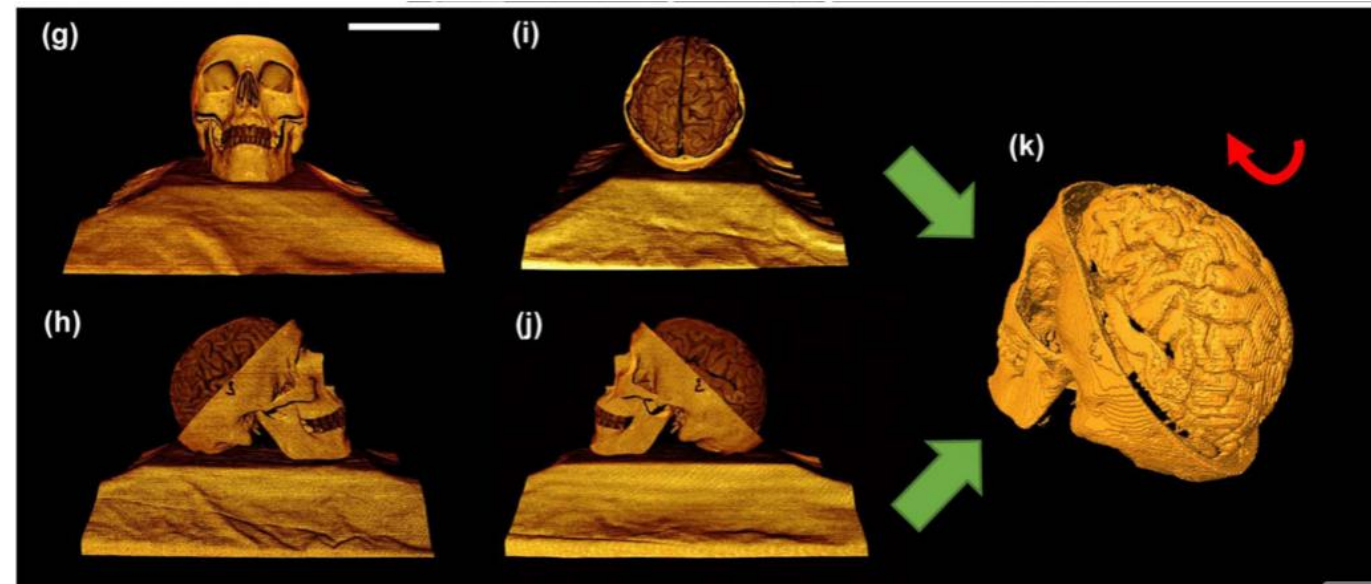
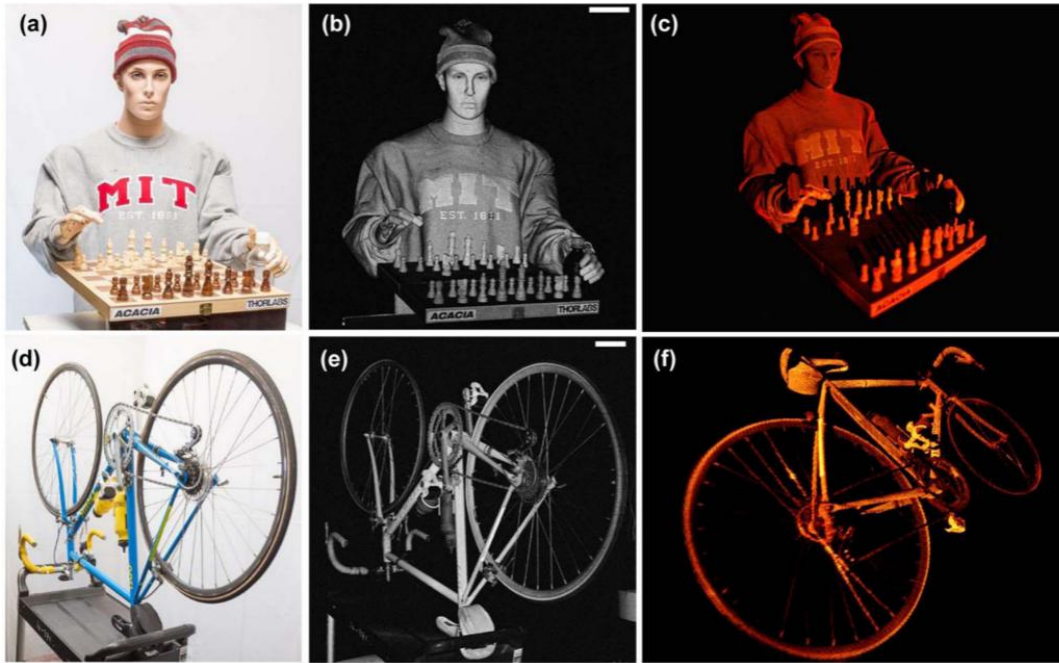
OCT IN AEROSPACE ENGINEERING: CHALLENGES

Portability is still a major drawback of this technique, which hinders its application in real world scenarios.



OCT IN AEROSPACE ENGINEERING: CHALLENGES

The imaging range of Fourier-domain OCT has been limited to a few centimeters. However, although it involves a more complex setup macroscopic imaging is possible. A group has achieved 1.5 meters!



<http://dx.doi.org/10.1364/OPTICA.3.001496>